

Differentially Private Preference Data Synthesis for Large Language Model Alignment

Anonymous Authors¹

Abstract

Preference alignment is a central post-training mechanism for large language models (LLMs), yet its reliance on real human preference data exposes sensitive prompts, responses, and human judgments. We introduce **DPPrefSyn**, a differentially private (DP) algorithm for synthesizing preference data that supports privacy-preserving LLM alignment. DPPrefSyn provides a principled synthesis pipeline grounded in the Bradley–Terry preference model and in the geometric structure of pairwise preference representations. The algorithm first learns DP-protected preference models from private comparisons, and then combines these models with public prompts to construct synthetic preference pairs. Its design captures heterogeneous human preferences through the shared linear structure of per-cluster reward models and improves private learning accuracy through DP Principal Component Analysis (DP-PCA). Experiments across question answering and summarization benchmarks show that DPPrefSyn delivers competitive alignment utility under strong DP guarantees, establishing DP synthetic preference data as a practical substrate for privacy-preserving alignment across multiple post-training methods. To the best of our knowledge, this is the first work to generate DP synthetic preference data for LLM alignment. Our code will be released upon acceptance.

1. Introduction

Preference alignment methods, including RLHF (Stiennon et al., 2020) and DPO (Rafailov et al., 2024), have become a standard mechanism for adapting LLMs to human expectations. These methods depend on pairwise preference data: annotators compare two candidate responses to the same

prompt and identify the response that better satisfies the intended criterion. The resulting comparisons guide post-training by increasing the model’s tendency to rank preferred responses above less preferred alternatives (Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022). Because this paradigm has proved effective for improving interactive model behavior, it now underlies deployed and research systems for chat assistance (Achiam et al., 2023), mathematical reasoning (Shao et al., 2024), and code generation (Shen et al., 2023).

The same data that makes preference alignment effective also creates a significant privacy challenge. Preference datasets frequently contain real user prompts and human judgments; prompts may reveal information about health, identity, or other sensitive topics, while feedback can expose private beliefs, preferences, and behavioral patterns (Li et al., 2023; Yu et al., 2024). Differential privacy (DP) (Dwork et al., 2006b) offers a rigorous framework for mitigating these risks, and existing DP alignment studies have produced encouraging progress (Chowdhury et al., 2024; Yu et al., 2024; Wu et al., 2023a). Yet the current landscape remains incomplete. Several approaches privatize only one component of the preference record, such as user prompts (Yu et al., 2024) or preference labels (Chowdhury et al., 2024; Zhang et al., 2025), leaving the remaining components available for inference. Other methods are specialized to particular post-training procedures, including RLHF (Wu et al., 2023a), and therefore do not directly support newer optimization methods such as DPO. Finally, methods that depend directly on limited private preference data inherit the cost and scalability constraints of human annotation, which can restrict alignment utility under strong privacy budgets. These limitations motivate the central question of this work:

Is it possible to achieve superior preference alignment while ensuring thorough privacy protection and maintaining compatibility with emerging alignment methodologies?

This paper answers that question affirmatively by constructing DP synthetic preference data that protects the private preference dataset while remaining usable by downstream alignment algorithms. The contributions are as follows:

- We introduce DPPrefSyn (Figure 1), an algorithm for gen-

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

erating DP synthetic preference data for LLM alignment. DPPrefSyn is grounded in the Bradley–Terry preference model and in the geometry of pairwise preference embeddings. Its design has three coordinated components: 1) *Capturing diverse human preferences via shared linear structure of rewards and clustering*. The Bradley–Terry model and a linear reward parameterization allow DPPrefSyn to cluster preference samples by embedding differences between prompt–preferred and prompt–dispreferred pairs, and to represent each cluster with a shared reward function. 2) *Improving learning efficiency through DP-PCA*. DP-PCA projects high-dimensional embeddings into a lower-dimensional subspace that preserves the preference signal needed for private clustering and reward learning. 3) *Saving privacy budgets by using public prompts*. Public prompts are used to elicit candidate responses, while DP reward models select synthetic preference pairs, concentrating the privacy budget on private preference modeling rather than prompt generation.

- We provide a privacy analysis establishing that DPPrefSyn satisfies (ϵ, δ) -DP. The analysis allocates privacy budget ϵ_0 to DP-PCA (Amin et al., 2019), ϵ_1 to DP-KMeans (Su et al., 2016), and the remaining budget to per-cluster reward-model training with DP-SGD. We use the Privacy Random Variable (PRV) accountant (Gopi et al., 2021) to compose the DP-SGD cost tightly. By DP post-processing, the synthesized preference dataset can subsequently be reused with multiple alignment methods and LLMs without additional privacy expenditure.
- We evaluate DPPrefSyn on OpenAssistant (Köpf et al., 2023) and Anthropic-HH (Bai et al., 2022) question answering tasks and on TL;DR summarization (Stiennon et al., 2020). Across these settings, DPPrefSyn produces DP synthetic preference data that maintains strong utility under privacy constraints. For example, at $\epsilon = 2$ on Anthropic-HH, DPO fine-tuning of Pythia-2.8B with DPPrefSyn data obtains a GPT-4o win rate of 56.48%, compared with 37.02% for fine-tuning on real data. The results demonstrate that DP synthetic preference data can support effective alignment under strong privacy guarantees and can be used with both DPO and RLHF-style downstream alignment procedures.

2. Related Work

Differentially Private Alignment of LLMs. Wu et al. (2023a) introduce a DP framework for reinforcement-learning-based LLM alignment by adapting PPO to the DP setting. That approach provides an important privacy mechanism for PPO, but it remains *limited to PPO* even as preference alignment increasingly relies on alternative objectives. Chowdhury et al. (2024) study reward estimation from preference-based feedback under label-DP (Ghazi

et al., 2021), and Zhang et al. (2025) propose AUP-RLHF as a user-level label-DP framework for RLHF. These methods protect preference labels, but they *do not address the sensitivity of prompts or responses themselves*. They also remain *constrained by the limited private user preference data* available for alignment, since human preference annotations are costly to collect at scale. Yu et al. (2024) instead protect sensitive user *instructions* by generating DP synthetic *instruction* data for supervised fine-tuning, but preference labels remain outside the privacy target. DPPrefSyn addresses the full preference record—prompts, responses, and labels—by synthesizing a DP preference dataset from private user data. Because the output is a DP dataset rather than a method-specific model update, it can be reused across LLMs and preference alignment methods for privacy-preserving annotation expansion and model fine-tuning.

Beyond preference alignment, a separate line of work studies private LLM fine-tuning with general labeled data rather than pairwise comparisons (Yu et al., 2021; Li et al., 2021; Chen et al., 2024; Tang et al., 2024; Zhang et al., 2023; He et al., 2022). Recent studies also protect in-context prompts used for in-context learning (Duan et al., 2023; Tang et al., 2023; Gao et al., 2024; Hong et al., 2023; Wu et al., 2023b). These settings are complementary to the preference-data synthesis problem addressed here.

Differentially Private Synthetic Text Generation. DP-PrefSyn belongs to the broader literature on DP synthetic text generation, where two representative paradigms have emerged. The first privately fine-tunes a language model on sensitive data and then samples synthetic text from the DP-trained model (Yue et al., 2023; Mattern et al., 2022; Mireshghallah et al., 2023; Carranza et al., 2023; Yu et al., 2024; Wang et al., 2024; Ochs & Habernal, 2024; Tan et al., 2025; Carranza et al., 2024). For example, Yu et al. (2024) privately fine-tune LLaMA on private instruction data and use the resulting model to generate synthetic text for supervised instruction tuning.

Because DP fine-tuning can be computationally expensive or inaccessible for non-public language models, a second paradigm generates DP synthetic data through LLM APIs (Lin et al., 2023; Xie et al., 2024; Wu et al., 2024; Hou et al., 2024). These methods typically prompt an LLM to generate candidates and then select outputs that resemble private data through a privacy-preserving selection procedure. For instance, Xie et al. (2024) propose Aug-PE, which iteratively refines LLM-generated samples using embedding-space similarity to private data. *DPPrefSyn extends this API-based synthetic-data perspective to preference alignment, where the object to be protected and reproduced is a structured pairwise preference comparison rather than a single text sample.*

Diversity in Human Preferences. Human preference data

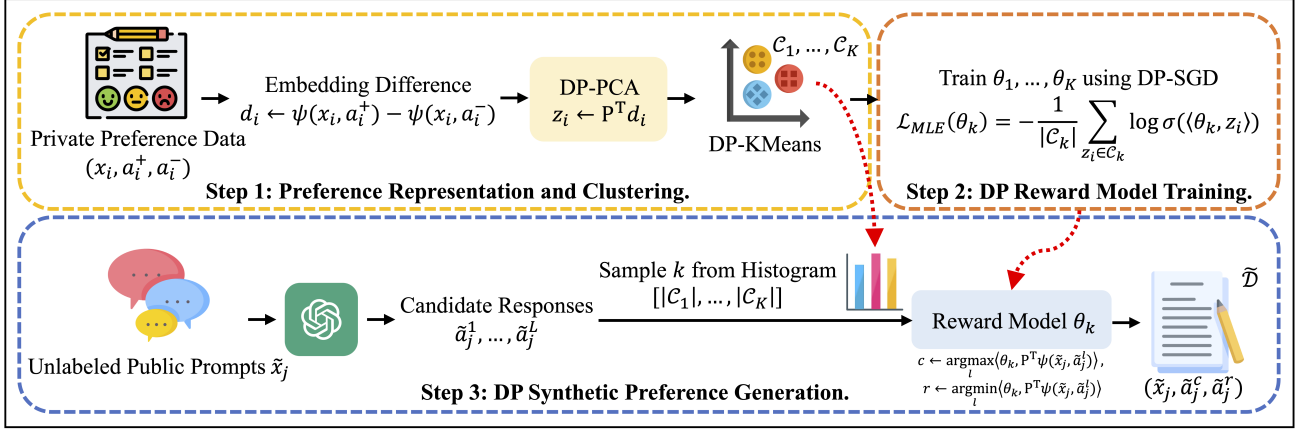


Figure 1. Overview of DPPrefSyn. The algorithm synthesizes DP preference data in three stages: 1) representing private comparisons as embedding differences and clustering them with DP-PCA and DP-KMeans, 2) training a DP reward model for each cluster with DP-SGD, and 3) generating synthetic preference pairs from public prompts using the DP-protected distribution over learned reward functions.

is not monolithic: preferences vary across annotators and user groups (Denton et al., 2021; Aroyo et al., 2023a;b; Chakraborty et al., 2024). Relevant sources of variation include sociodemographic differences, personal beliefs and biases, domain expertise, and ambiguity in natural language interpretation (Sandri et al., 2023; Vogels, 2021). This observation has motivated preference-alignment methods that adapt to different user values or groups (Lee et al., 2024b; Poddar et al., 2024; Singh et al., 2025). For example, Singh et al. (2025) propose a meta-learning framework in which an LLM adapts to individual users from a small number of preference examples. DPPrefSyn incorporates this heterogeneity directly into DP synthesis: it generates DP synthetic preference data that *captures the underlying diversity of human feedback*, yielding stronger alignment performance than variants that ignore such diversity (Section 5.3).

3. Problem Definition

Preference Alignment of LLMs. We study LLM preference alignment from pairwise human feedback, represented by a preference dataset $\mathcal{D} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$. Each record contains a prompt x_i and two candidate responses a_i^+, a_i^- , where human annotators prefer a_i^+ to a_i^- .

Following prior works on reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) (Ouyang et al., 2022; Zhu et al., 2023; Rafailov et al., 2024; Liu et al., 2023), we model human preferences using the Bradley–Terry model (Bradley & Terry, 1952):

$$\mathbb{P}[a_i^+ \succ a_i^- | x_i] = \frac{\exp(r^*(x_i, a_i^+))}{\exp(r^*(x_i, a_i^+)) + \exp(r^*(x_i, a_i^-))} = \sigma(r^*(x_i, a_i^+) - r^*(x_i, a_i^-)),$$

where $a_i^+ \succ a_i^-$ denotes the preference of a_i^+ over a_i^- , $r^*(x, a)$ is the latent reward function, and $\sigma(z) = 1/(1 +$

$e^{-z})$ is the sigmoid function.

This probabilistic formulation underlies both RLHF, which explicitly estimates a reward model from preference data, and DPO, which induces an implicit reward through the log-ratio between the learned policy and a reference policy.

Differentially Private Preference Data Synthesis. Our objective is to protect the private preference dataset $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ from adversaries who attempt to infer information about individual prompts, responses, or preference labels from the outputs of an aligned LLM. We therefore synthesize a DP preference dataset $\tilde{\mathcal{D}} = \{(\tilde{x}_j, \tilde{a}_j^+, \tilde{a}_j^-)\}_{j=1}^m$ that retains the preference structure needed for alignment while satisfying formal privacy constraints. By the post-processing property of DP, $\tilde{\mathcal{D}}$ can be reused across preference alignment methods and LLMs without additional privacy cost.

We use the standard definition of (ϵ, δ) -differential privacy.

Definition 3.1 ((ϵ, δ) -Differential Privacy (Dwork et al., 2006a)). A randomized algorithm $\mathcal{A}$ is (ϵ, δ) -differentially private if for any two neighboring inputs $\mathcal{D}$ and $\mathcal{D}'$ that differ by a single entry and any set $\mathcal{S}$ of possible outputs: $\mathbb{P}[\mathcal{A}(\mathcal{D}) \in \mathcal{S}] \leq e^\epsilon \mathbb{P}[\mathcal{A}(\mathcal{D}') \in \mathcal{S}] + \delta$.

Diverse User Preferences and Linear Reward Structure. Human preferences in a private dataset are typically heterogeneous rather than homogeneous (Chakraborty et al., 2024; Bakker et al., 2022; Kovač et al., 2023; Jang et al., 2023; Rame et al., 2023; Ji et al., 2023). Annotators may emphasize different response attributes, including factual accuracy, politeness, creativity, and clarity, thereby producing diverse and sometimes conflicting preference signals. Effective synthesis of $\tilde{\mathcal{D}}$ must therefore preserve the preference diversity present in $\mathcal{D}_{\text{priv}}$.

To model this diversity in a tractable form, we assume a

linear structure for the latent reward function (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023). Specifically, we adopt a linear reward model

$$r_\theta(x, a) = \langle \theta, \phi(x, a) \rangle,$$

where $\phi(x, a) \in \mathbb{R}^d$ is the *shared* feature vector and $\theta \in \mathbb{R}^d$ is a *user-dependent* parameter.

This linear structure balances *expressiveness* and *tractability* under privacy constraints. Although individual users may express heterogeneous preferences, their reward functions can share semantic features in the embedding representation $\phi(x, a)$ while differing in the weights encoded by θ . The parameterization therefore captures common structure across responses while permitting variation across users or preference clusters, which is essential when each private subgroup provides limited data and each learning step consumes privacy budget. The empirical comparison in Appendix B.6 confirms that this linear reward structure captures useful preference signals and remains competitive with a fully fine-tuned GPT-2 reward model despite its simplicity.

4. Proposed Method

4.1. Key Components of the Algorithm Design

Before presenting the full algorithm, we identify the design components that make DP preference synthesis accurate, privacy-efficient, and robust to heterogeneous feedback.

Leveraging Linear Reward Structure for Preference Clustering. The linear reward model implies that

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \sigma(\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle).$$

For a comparison tuple (x_i, a_i^+, a_i^-) , the event $a_i^+ \succ a_i^-$ corresponds to a large positive value of $\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle$, so the preference parameter θ is aligned with the feature difference $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$. Comparisons generated by annotators with similar preferences should therefore induce feature-difference vectors pointing in related directions in the embedding space. DPPrefSyn uses this structure to cluster preference samples by their difference vectors, so that comparisons assigned to the same cluster can be approximated by a shared reward function with a common preference vector θ . Appendix B.9 shows that the resulting clusters are meaningful and interpretable in terms of preference styles.

Adopting DP-PCA to Obtain Effective Features. A natural instantiation of $\phi(x, a)$ is an embedding of the corresponding prompt–response pair, since such embeddings encode semantic and stylistic properties that shape human preference judgments. These embeddings are typically high-dimensional (e.g., 1024 dimensions), whereas each preference cluster contains only a subset of the private data.

Direct private learning in the original embedding space is therefore sample-inefficient and can destabilize clustering. DPPrefSyn applies differentially private PCA (DP-PCA) (Amin et al., 2019; Liu et al., 2022) to obtain compact feature representations for both clustering and reward-model training, improving sample efficiency while retaining the preference information required for alignment.

Utilizing Public Prompts to Save DP Budget. Jointly synthesizing prompts, preferred responses, and less preferred responses would spend privacy budget on all components of the preference tuple and can substantially reduce utility. DPPrefSyn instead uses public prompts, allowing the privacy budget to be concentrated on learning the private preference structure over responses. For each public prompt, an LLM generates multiple candidate responses; DP-trained reward functions then select response pairs that instantiate the learned preference distribution. Although public and private prompts may not follow identical distributions, the experiments show that the resulting synthetic data preserves the statistical patterns of the private preferences sufficiently to support effective alignment.

4.2. DPPrefSyn Algorithm

The complete pipeline is summarized in Figure 1 and Algorithm 1. DPPrefSyn proceeds through three stages: (1) DP preference clustering induced by the linear reward structure, (2) DP training of a reward model within each cluster, and (3) synthetic preference generation from public prompts using the learned DP reward models.

Step 1: DP Preference Clustering. For each private comparison (x_i, a_i^+, a_i^-) , DPPrefSyn first computes the feature difference vector $d_i = \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ (Line 3), where $\psi(x, a)$ is an embedding model that maps a prompt–response pair to a d -dimensional vector.

The algorithm then applies DP-PCA (Amin et al., 2019) (Line 5) and obtains the projected feature difference $z_i = \mathbf{P}^\top d_i$ (Line 6), where $\mathbf{P} \in \mathbb{R}^{d \times p}$ is the DP-PCA projection matrix. In effect, DP-PCA (Amin et al., 2019) privately approximates the dominant covariance directions and preserves the principal preference-related structure while reducing the dimensionality that governs private clustering and reward learning.

After projection, DPPrefSyn applies DP-KMeans (Su et al., 2016) to $\{z_i\}_{i=1}^n$ and obtains K preference clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ (Line 7). We use the algorithm of (Su et al., 2016), a DP variant of Lloyd’s algorithm.

Step 2: DP Reward Model Training. For each cluster $\mathcal{C}_k$, DPPrefSyn estimates a shared preference vector $\theta_k \in \mathbb{R}^p$

by minimizing the negative log-likelihood (Line 9):

$$\mathcal{L}_{\text{MLE}}(\theta_k) = -\frac{1}{|C_k|} \sum_{z_i \in C_k} \log \sigma(\langle \theta_k, z_i \rangle), \quad (1)$$

where z_i is the DP-PCA-projected embedding difference from Step 1. Each cluster-specific reward model is trained with DP-SGD (Abadi et al., 2016) using noise multiplier σ_2 .

Step 3: DP Synthetic Preference Generation. DPPrefSyn computes the cluster-count histogram $\mathbf{h} = [|C_1|, \dots, |C_K|]$ (Line 11) and normalizes it to obtain the cluster sampling distribution $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ (Line 12).

For each public prompt $\tilde{x}_j$, a relatively high-temperature LLM call produces L diverse candidate responses $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ (Line 14). The algorithm samples a cluster index $k \sim \mathbf{p}$ from the DP histogram (Line 15) and scores the candidates with the corresponding reward model θ_k . Specifically, it embeds each candidate pair as $\psi(\tilde{x}_j, \tilde{a}_j^l)$, projects the embedding with $\mathbf{P}$, and assigns reward $\langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$. The highest- and lowest-scoring candidates become the preferred and less preferred responses: $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$, $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ (Lines 16-17). When the reward gap is too small (e.g., < 0.5), DPPrefSyn discards the sample to maintain preference quality; otherwise, it appends $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$ to $\tilde{\mathcal{D}}$. Repeating this procedure over all public prompts yields the DP synthetic preference dataset $\tilde{\mathcal{D}}$.

Privacy Analysis of DPPrefSyn. The privacy budget is divided across the private operations: ε_0 is assigned to DP-PCA (Line 5), ε_1 is assigned to DP-KMeans (Line 7), and the remaining budget is used to train per-cluster reward models with DP-SGD and noise multiplier σ_2 (Line 9). For the DP reward models, we invoke parallel composition: because the clusters are disjoint, changing one training record affects only one cluster, so the aggregate privacy cost is controlled by the cost of training on the smallest retained cluster. Clusters with too few samples are discarded to ensure a known lower bound on cluster size. The Privacy Random Variable (PRV) accountant (Gopi et al., 2021) composes the per-step DP-SGD privacy losses, yielding $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP for reward-model training. All remaining operations use public data or constitute post-processing and therefore introduce no additional privacy cost. The full proof and privacy-accounting code are provided in Appendix A.

Discussion on Prompt Distribution Shift. DPPrefSyn is designed to remain useful when public prompts are not drawn from exactly the same distribution as the private prompts. The key observation is that preference diversity is represented by θ_k , the linear reward parameter, rather than by $\phi(x, a)$ alone, which encodes a particular prompt-response pair. A user may issue many kinds of prompts

Algorithm 1 DPPrefSyn

```

1: Input: Private dataset  $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ , public prompts  $\{\tilde{x}_j\}_{j=1}^m$ , embedding model  $\psi$ , number of clusters  $K$ , an LLM  $\text{LLM}(\cdot)$ , DP parameters  $\varepsilon_0, \varepsilon_1, \sigma_2$ .
2: for each  $(x_i, a_i^+, a_i^-)$  in  $\mathcal{D}_{\text{priv}}$  do
3:    $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ , where  $d_i \in \mathbb{R}^d$ 
4: end for
5:  $\mathbf{P} \leftarrow \text{DP-PCA}(\{d_i\}_{i=1}^n, \varepsilon_0)$ , where  $\mathbf{P} \in \mathbb{R}^{d \times p}$ 
6:  $\{z_i\}_{i=1}^n \leftarrow \{\mathbf{P}^\top d_i\}_{i=1}^n$ 
7:  $C_1, \dots, C_K \leftarrow \text{DP-KMeans}(\{z_i\}_{i=1}^n, K, \varepsilon_1)$ 
8: for each cluster  $C_k$  do
9:   Train a linear reward model  $\theta_k$  on  $C_k$  using DP-SGD with noise multiplier  $\sigma_2$  (Equation (1))
10: end for
11:  $\mathbf{h} \leftarrow [|C_1|, \dots, |C_K|]$ 
12:  $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ 
13: for each public prompt  $\tilde{x}_j$  do
14:   Generate responses  $\tilde{a}_j^1, \dots, \tilde{a}_j^L \leftarrow \text{LLM}(\tilde{x}_j)$ 
15:   Sample cluster index  $k \sim \mathbf{p}$ 
16:    $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
17:    $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
18:   Add  $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$  to  $\tilde{\mathcal{D}}$ 
19: end for
20: return  $\tilde{\mathcal{D}}$ .
```

while maintaining a relatively stable underlying preference profile. Accordingly, if the distribution over learned parameters θ_k reflects the distribution in the private dataset, synthetic comparisons generated on public prompts can still express the private preference patterns. DPPrefSyn enforces this correspondence by sampling a cluster index k according to the DP histogram (Line 15) for each public prompt and then selecting response pairs with the corresponding reward model. Thus, even under prompt distribution shift, the synthetic data can preserve the preference structure learned from the private data.

5. Experiments

Datasets. We evaluate on question answering and summarization tasks. For question answering, we use the OpenAssistant dataset (Köpf et al., 2023), which contains assistant-style conversations, and the Anthropic-HH dataset (Bai et al., 2022), which provides human preference comparisons focused on helpfulness and harmlessness. For summarization, we use the TL;DR dataset (Stiennon et al., 2020), which contains annotations of human preference on pairs of summaries. To simulate public data, we use prompts from Alpaca (Taori et al., 2023) for the OpenAssistant QA task, SafeRLHF (Ji et al., 2024) for the Anthropic-HH QA task, and XSum (Narayan et al., 2018) for the TL;DR summarization task. Additional details are available in Appendix B.2.

Setup for Algorithm 1. By default, we use BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ . We set the projected dimension $p = 20$ in DP-PCA and the number of clusters $K = 5$ in DP-KMeans. For DP-SGD, we use a learning rate of 0.1, a batch size of 4, 4 training epochs, and a gradient clipping norm of 1.0. Candidate responses are generated using the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) with a temperature of 0.9. We set the number of candidates per prompt to $L = 5$. We consider overall privacy budgets $\varepsilon = 0.5, 1, 2, 4, 8$ and set $\delta = 1/|\mathcal{D}_{\text{priv}}|$. We study the effect of different generator models in Table 3, embedding models and hyperparameters in Section 5.3.

Evaluation. For downstream preference alignment, we fine-tune the Pythia-2.8B model (Biderman et al., 2023). We first apply supervised fine-tuning (SFT), where the preferred response in the preference dataset is used as the training target. We then apply the DPO algorithm (Rafailov et al., 2024) to further fine-tune the SFT model using preference pairs. Following Rafailov et al. (2024), we measure the win rate of the model-generated responses against the preferred responses in the test set using the GPT-4o model. We study more types of downstream models, and different preference alignment methods as ablation study in Section 5.2.

Baselines. We compare DPPrefSyn with fine-tuning the downstream model on real data using DP-SGD (Abadi et al., 2016), denoted as DP-FT. Note that DPPrefSyn is more reusable and versatile than DP-FT because it generates DP synthetic data that can be used to train different models with different alignment algorithms without incurring additional privacy cost, while DP-FT consumes the privacy budget in training a single privatized model. We also include a fully private baseline $\varepsilon = 0$, where we evaluate the base model without any preference fine-tuning.

Details about the setups, hyperparameters, metrics, and baselines are provided in Appendix B.3.

5.1. Understanding the Performance of DPPrefSyn

In this section, we analyze the performance of DPPrefSyn by answering two research questions about its privacy-utility trade-off. Our main results are presented in Table 1.

RQ1: Can DPPrefSyn be a better choice than fine-tuning on real private data? DPPrefSyn consistently achieves a stronger privacy-utility trade-off than DP-FT across all tasks under $\varepsilon = 0.5, 1, 2, 4, 8$, and outperforms the utility of fine-tuning on real private data without DP constraints (DP-FT with $\varepsilon = \infty$), while preserving privacy. For instance, on the OpenAssistant QA task with $\varepsilon = 2$, our approach achieves 9.10% after SFT and 11.04% after DPO, compared to DP-FT’s 4.38% after SFT and 6.18% after DPO. Remarkably, while still preserving privacy, DP-

PrefSyn achieves higher utility than the non-private baseline of fine-tuning on real private data (DP-FT with $\varepsilon = \infty$), which reaches 4.75% after SFT and 8.20% after DPO. Similarly, on the Anthropic-HH dataset with $\varepsilon = 2$, DPPrefSyn achieves a 55.21% win rate after SFT and 56.48% after DPO, compared to 31.53% and 37.02% achieved by DP-FT, respectively. It also outperforms direct fine-tuning on real data (DP-FT with $\varepsilon = \infty$), which achieves 31.98% after SFT and 38.72% after DPO.

These results highlight two **key advantages** of our DP synthetic data. **First**, DPPrefSyn leads to significant performance improvement after SFT alone compared to directly applying SFT on private data. For example, on the TL;DR summarization task with $\varepsilon = 2$, DPPrefSyn achieves a win rate of 62.49% after SFT, much higher than the 22.57% achieved by SFT on private data. This is mainly because DPPrefSyn generates candidate responses using a high-performance LLM, leading to high-quality training data that provides more effective supervision during SFT. This advantage holds even with limited public data. For instance, on the OpenAssistant QA task with $\varepsilon = 4$ and only 5K public prompts, DPPrefSyn achieves a win rate of 6.80% after SFT, compared to 4.46% for SFT directly on private data (Table 4), which contains around 14K samples, highlighting DPPrefSyn’s ability to handle data scarcity while preserving utility. **Second**, applying DPO with our synthetic data on top of the SFT model leads to further improvements, showing that our synthetic preference pairs capture meaningful distinctions aligned with human preferences. This is enabled by the private reward functions that capture diverse user preferences while preserving privacy. Compared to directly using private data, the improvement from DPO is smaller in our setting, likely because the SFT model trained on our synthetic data already performs well, leaving less room for optimization. However, for the entire fine-tuning process, the performance gain using our DP synthetic dataset is still significantly higher than that using real private data.

RQ2: How does DPPrefSyn perform across different privacy budget ε ? First, Table 1 shows that DPPrefSyn generally achieves better performance compared to DP-FT as ε increases from 0.5, 1, 2, 4, 8 to ∞ , indicating good scalability with privacy budget ε . **Second**, DPPrefSyn can be more robust to DP noise than DP-FT in some cases. For example, on OpenAssistant with Llama-3-8B as the downstream model, DPO performance for DP-FT drops from 59.13% to 45.79% when ε decreases from ∞ to 4, while DPPrefSyn drops only slightly from 61.52% to 60.25% (see Table 3 in Section 5.2). The reason could be that DP-FT perturbs model parameters directly through DP-SGD, whereas DPPrefSyn adds noise during data synthesis, which better preserves utility.

Table 1. GPT-4o win rates on test sets with the downstream Pythia-2.8B model fine-tuned by SFT and DPO. $\varepsilon = 0$ denotes the base (fully private, non-finetuned) LLM. Compared with DP-FT, DPPrefSyn achieves higher win rates under privacy levels $\varepsilon = 0.5, 1, 2, 4, 8$, even outperforming the utility of non-private fine-tuning (DP-FT with $\varepsilon = \infty$). Results show mean and standard deviation over 5 random seeds.

Task	$\varepsilon = 0$	Method	Data Type	Pref. Alignment	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	$\varepsilon = \infty$
OpenAssistant	2.11	DP-FT	Original	SFT	3.96 _{0.20}	4.29 _{0.21}	4.38 _{0.23}	4.46 _{0.18}	4.49 _{0.30}	4.75 _{0.27}
				SFT+DPO	5.65 _{0.32}	6.04 _{0.58}	6.18 _{0.83}	6.12 _{0.63}	6.32 _{0.71}	8.20 _{0.62}
		DPPrefSyn	Synthetic	SFT	8.96 _{0.39}	9.07 _{0.32}	9.10 _{0.51}	9.33 _{0.56}	9.63 _{0.38}	9.80 _{0.58}
				SFT+DPO	9.86_{0.59}	10.31_{0.54}	11.04_{0.92}	11.85_{0.99}	12.05_{0.82}	12.36_{0.55}
Anthropic-HH	12.14	DP-FT	Original	SFT	29.77 _{0.76}	31.10 _{0.72}	31.53 _{0.77}	31.29 _{0.65}	31.37 _{0.67}	31.98 _{0.56}
				SFT+DPO	35.00 _{0.49}	36.27 _{0.63}	37.02 _{0.72}	36.74 _{0.90}	36.94 _{1.06}	38.72 _{0.56}
		DPPrefSyn	Synthetic	SFT	54.90 _{0.66}	55.19 _{0.91}	55.21 _{0.82}	54.95 _{1.10}	55.63 _{0.86}	55.95 _{0.70}
				SFT+DPO	55.08_{1.04}	55.96_{0.80}	56.48_{0.61}	56.51_{1.93}	56.86_{1.17}	57.53_{0.93}
TL;DR	11.64	DP-FT	Original	SFT	21.19 _{1.12}	22.41 _{1.09}	22.57 _{0.65}	22.45 _{0.90}	23.04 _{0.79}	23.82 _{1.26}
				SFT+DPO	59.69 _{0.82}	62.21 _{1.12}	62.02 _{1.67}	63.00 _{0.73}	63.06 _{0.84}	64.52 _{1.79}
		DPPrefSyn	Synthetic	SFT	61.92 _{1.06}	62.14 _{1.61}	62.49 _{1.48}	62.72 _{1.66}	62.29 _{1.14}	62.55 _{1.23}
				SFT+DPO	67.50_{0.79}	68.07_{1.95}	68.54_{2.76}	68.56_{1.59}	71.01_{2.17}	72.20_{1.08}

Table 2. DPPrefSyn achieves higher win rates than fine-tuning on private data across SFT, DPO, and PPO, while providing strong privacy guarantees on OpenAssistant. Results show mean and standard deviation over 5 random seeds.

Pref. Alignment	$\varepsilon = 0$	DPPrefSyn ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
SFT		9.33_{0.56}	4.75 _{0.27}
SFT+DPO	2.11	11.85_{0.99}	8.20 _{0.62}
SFT+PPO		10.11_{0.61}	7.08 _{0.78}

5.2. Understanding the Properties of DPPrefSyn

In this section, we study properties of DPPrefSyn, including its generalizability across different preference alignment methods and LLMs, its performance under varying scales of public data, and its robustness to empirical privacy attacks.

RQ3: Is DPPrefSyn effective across different preference alignment methods? DPPrefSyn supports various preference alignment methods and remains effective under strong DP constraints. In Table 2, we evaluate DPPrefSyn with SFT, DPO, and PPO. For PPO, we train K reward models (one for each cluster) to capture diverse preference patterns. During PPO training, we use the average output of the K reward models as the reward and optimize the policy. We find that DPPrefSyn consistently achieves better utility for SFT, DPO, and PPO, compared with fine-tuning on private data, while preserving strong privacy guarantees.

RQ4: Can more powerful downstream LLMs benefit from synthetic data generated by DPPrefSyn? DPPrefSyn can be effective across a wide range of downstream LLMs. In Table 3, we conduct experiments on different downstream models, including Qwen-3-4B, Gemma-2-2B, and Llama-3-8B, using Llama-3-8B-Instruct and Qwen3-4B-Instruct-2507 as generator models. We find that DPPrefSyn can outperform fine-tuning on real data, while ensuring privacy across different model families after DPO. For example, using Qwen3-4B-Instruct-2507 as the generator, DPPrefSyn achieves an 86.52% win rate on Qwen-3-4B with $\varepsilon = 4$, which significantly outperforms non-private fine-tuning on real data

Table 3. DPPrefSyn achieves higher win rates than fine-tuning on private data across different downstream LLMs after DPO, while providing strong privacy guarantees on OpenAssistant. DPPrefSyn can also benefit from more powerful generator LLMs.

Downstream LLM	$\varepsilon = 0$	Method	Generator LLM	$\varepsilon = 4$	$\varepsilon = \infty$
Qwen-3-4B	48.17	DP-FT	-	54.92	59.41
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	86.52	87.08
			Llama-3-8B-Instruct	51.69	52.11
Gemma-2-2B	7.30	DP-FT	-	30.06	33.85
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	36.24	37.08
			Llama-3-8B-Instruct	36.01	36.66
Llama-3-8B	16.15	DP-FT	-	45.79	59.13
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	60.25	61.52
			Llama-3-8B-Instruct	42.28	44.10

Table 4. DPPrefSyn achieves higher win rates than fine-tuning on private data with varying public data size m . Results show mean and standard deviation over 5 random seeds.

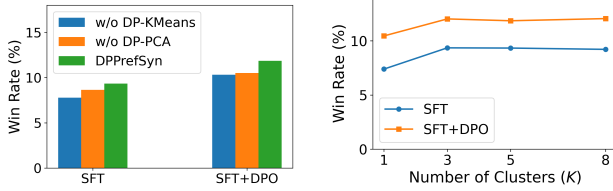
	DPPrefSyn ($\varepsilon = 4$)			DP-FT ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
Data Type (Size)	Synthetic ($m = 5,000$)	Synthetic ($m = 10,000$)	Synthetic ($m = 52,000$)	Original (14,167)	Original (14,167)
SFT	6.80 _{0.66}	7.30 _{0.48}	9.33 _{0.56}	4.46 _{0.18}	4.75 _{0.27}
SFT+DPO	9.44 _{0.45}	9.55 _{0.92}	11.85 _{0.99}	6.12 _{0.63}	8.20 _{0.62}

(59.41%). We also observe that DPPrefSyn’s performance improves with stronger generator models, with Qwen3-4B-Instruct-2507 outperforming Llama-3-8B-Instruct.

RQ5: Can DPPrefSyn be effective with varying scales of public data? We study the effect of public data size m in Table 4. The private dataset (OpenAssistant) contains 14K preference annotations, while the full public dataset (Alpaca) contains 52K prompts. We randomly sample subsets of the public prompts at different sizes ($m = 5K, 10K, 52K$) to generate synthetic preference data using DPPrefSyn. We find that even with only 5K public prompts, DPPrefSyn achieves a win rate of 6.80% after SFT and 9.44% after DPO under $\varepsilon = 4$, outperforming the DP-FT baseline, which achieves 4.46% and 6.12%, respectively. This shows that **DPPrefSyn remains effective even when the available public data is limited**, thanks to the high-quality responses

Table 5. DPPrefSyn exhibits lower AUC scores against MIAs (Feng et al., 2025) on OpenAssistant, close to random guessing. Results show mean and standard deviation over 5 random seeds.

Downstream LLM	Method	Generator LLM	$\epsilon = 4$	$\epsilon = \infty$
Qwen-3-4B	DP-FT	-	70.15 _{2.09}	73.52 _{2.30}
	DPPrefSyn	Qwen-3-4B-Instruct	50.99 _{1.10}	50.87 _{0.84}
		Llama-3-8B-Instruct	50.64 _{1.29}	49.98 _{1.19}
Gemma-2-2B	DP-FT	-	62.88 _{1.78}	65.39 _{2.59}
	DPPrefSyn	Qwen-3-4B-Instruct	50.51 _{1.38}	50.43 _{0.91}
		Llama-3-8B-Instruct	50.77 _{0.83}	51.10 _{0.54}
Llama-3-8B	DP-FT	-	61.95 _{2.05}	70.13 _{1.24}
	DPPrefSyn	Qwen-3-4B-Instruct	50.53 _{0.62}	50.96 _{0.95}
		Llama-3-8B-Instruct	50.45 _{0.87}	50.67 _{0.96}



(a) Removing DP-KMeans or DP-PCA.

(b) Varying clusters.

generated by the LLMs and the private reward functions that accurately capture user preferences. Besides, **as the public data size increases, the performance improvement over the baseline becomes more prominent**, indicating that DPPrefSyn can effectively leverage the vast amounts of public data for preference alignment, further mitigating the data scarcity issue in private preference data.

RQ6: Can DPPrefSyn effectively mitigate empirical privacy attacks? In Table 5, we perform membership inference attacks (MIAs) (Shokri et al., 2017) against the DPO fine-tuned downstream models on OpenAssistant and report the AUC score. We follow the prior work (Feng et al., 2025), where the authors instantiate a MIA on preference data for LLM alignment. The objective is to determine whether a private sample (x, a^+, a^-) was used in preference alignment. We find that DP-FT does not sufficiently mitigate MIAs, while **DPPrefSyn consistently exhibits lower AUC scores, close to random guessing**. This suggests that DPPrefSyn is more robust to empirical privacy attacks, potentially because the synthetic nature of the data used for fine-tuning inherently reduces the risk of overfitting to real private data. We defer the details to Appendix B.4.

5.3. Validating the Design of DPPrefSyn

We validate the design of DPPrefSyn by studying its algorithmic components in this section.

RQ7: How does modeling diverse human preferences impact DPPrefSyn’s performance? We evaluate this by removing either DP-KMeans or DP-PCA, and varying the number of clusters K , using OpenAssistant with $\epsilon = 4$. **(i) Removing either DP-KMeans or DP-PCA.** Figure 2a shows

that removing either DP-KMeans or DP-PCA leads to a performance drop. Without DP-KMeans, all samples are assigned to a single cluster. Removing DP-PCA leads to ineffective clustering and sample-inefficient learning in a high-dimensional space. Using multiple clusters with both components improves performance over using a single cluster. **(ii) Varying the number of clusters K .** In Figure 2b, we evaluate DPPrefSyn with $K = 1, 3, 5, 8$ clusters. As K increases, performance first improves at $K = 3$ and then remains relatively stable at $K = 5, 8$. This suggests that DPPrefSyn is robust to K , and moderate values (e.g., $K = 3, 5$) provide a good trade-off between diversity and data efficiency. Overall, these results show the benefit of capturing diverse preferences using multiple clusters.

RQ8: How do different embedding models or hyperparameters affect DPPrefSyn’s performance? We perform ablation studies on the embedding model ψ , DP-PCA dimension p , number of candidates per prompt L , filter threshold, and privacy budget allocation in Appendix B.5. Across different sentence encoders, including bge-large-en-v1.5 (Xiao et al., 2023), all-mpnet-base-v2 (Song et al., 2020), and all-distilroberta-v1 (Sanh et al., 2019), DPPrefSyn remains consistently effective. We find that $p = 20$ provides the best overall performance after DPO, $L = 5$ serves as a good default choice, and a moderate filtering threshold of 0.5 slightly outperforms no filtering, while a high threshold of 1 is too aggressive and hurts performance. Finally, DPPrefSyn is robust to reasonable changes in the budget allocation ϵ_0 and ϵ_1 .

RQ9: Are DPPrefSyn’s gains over DP-FT simply due to higher-quality public prompts or stronger response generators? To rule out confounding effects, we conduct controlled ablations where all methods share the same prompt set or response generator. Under these settings, DPPrefSyn continues to outperform DP-FT (Appendix B.7), showing that **its performance gains do not merely arise from higher-quality public prompts or stronger response generators**.

6. Conclusion

In this work, we introduced DPPrefSyn, an algorithm for synthesizing DP preference data from a private dataset. By modeling preference diversity via clustering, improving efficiency with DP-PCA, and conserving privacy budgets through public prompts, DPPrefSyn enables effective and private preference data synthesis for the alignment of LLMs. Our experiments show that DPPrefSyn achieves privacy guarantees while outperforming fine-tuning on real data.

7. Limitations

DPPrefSyn requires LLM API access to synthesize preference data, which may incur additional API costs. Our work

focuses on English text-based preference data, and extending DPPrefSyn to multilingual settings and multimodal data remains an open direction. Additionally, our experiments cover models up to 8B parameters due to computational constraints.

Impact statement

This work enables preference alignment of large language models without exposing personal information, which helps balance the benefits of big data with the need to protect individual privacy, promoting ethical data usage and fostering societal trust in an increasingly data-driven world.

References

- Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., and Zhang, L. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*, pp. 308–318, 2016.
- Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Amin, K., Dick, T., Kulesza, A., Munoz, A., and Vassilvitskii, S. Differentially private covariance estimation. *Advances in Neural Information Processing Systems*, 32, 2019.
- Aroyo, L., Diaz, M., Homan, C., Prabhakaran, V., Taylor, A., and Wang, D. The reasonable effectiveness of diverse evaluation data. *arXiv preprint arXiv:2301.09406*, 2023a.
- Aroyo, L., Taylor, A., Diaz, M., Homan, C., Parrish, A., Serapio-García, G., Prabhakaran, V., and Wang, D. Dices dataset: Diversity in conversational ai evaluation for safety. *Advances in Neural Information Processing Systems*, 36:53330–53342, 2023b.
- Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., Das-Sarma, N., Drain, D., Fort, S., Ganguli, D., Henighan, T., et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022.
- Bakker, M., Chadwick, M., Sheahan, H., Tessler, M., Campbell-Gillingham, L., Balaguer, J., McAleese, N., Glaese, A., Aslanides, J., Botvinick, M., et al. Fine-tuning language models to find agreement among humans with diverse preferences. *Advances in Neural Information Processing Systems*, 35:38176–38189, 2022.
- Biderman, S., Schoelkopf, H., Anthony, Q. G., Bradley, H., O’Brien, K., Hallahan, E., Khan, M. A., Purohit, S., Prashanth, U. S., Raff, E., et al. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning*, pp. 2397–2430. PMLR, 2023.
- Bradley, R. A. and Terry, M. E. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- Carranza, A., Farahani, R., Ponomareva, N., Kurakin, A., Jagielski, M., and Nasr, M. Synthetic query generation for privacy-preserving deep retrieval systems using differentially private language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 3920–3930, 2024.
- Carranza, A. G., Farahani, R., Ponomareva, N., Kurakin, A., Jagielski, M., and Nasr, M. Privacy-preserving recommender systems with synthetic query generation using differentially private large language models. *arXiv preprint arXiv:2305.05973*, 2023.
- Chakraborty, S., Qiu, J., Yuan, H., Koppel, A., Manocha, D., Huang, F., Bedi, A., and Wang, M. Maxmin-rlhf: Alignment with diverse human preferences. In *International Conference on Machine Learning*, pp. 6116–6135. PMLR, 2024.
- Chen, T., Da, L., Zhou, H., Li, P., Zhou, K., Chen, T., and Wei, H. Privacy-preserving fine-tuning of large language models through flatness. *arXiv preprint arXiv:2403.04124*, 2024.
- Chowdhury, S. R., Zhou, X., and Natarajan, N. Differentially private reward estimation with preference feedback. In *International Conference on Artificial Intelligence and Statistics*, pp. 4843–4851. PMLR, 2024.
- Denton, R., Díaz, M., Kivlichan, I., Prabhakaran, V., and Rosen, R. Whose ground truth? accounting for individual and collective identities underlying dataset annotation. *arXiv preprint arXiv:2112.04554*, 2021.
- Duan, H., Dziedzic, A., Papernot, N., and Boenisch, F. Flocks of stochastic parrots: Differentially private prompt learning for large language models. *Advances in Neural Information Processing Systems*, 36:76852–76871, 2023.
- Dwork, C., Kenthapadi, K., McSherry, F., Mironov, I., and Naor, M. Our data, ourselves: Privacy via distributed noise generation. In *Advances in Cryptology-EUROCRYPT 2006: 24th Annual International Conference on the Theory and Applications of Cryptographic Techniques, St. Petersburg, Russia, May 28-June 1, 2006. Proceedings 25*, pp. 486–503. Springer, 2006a.

- Dwork, C., McSherry, F., Nissim, K., and Smith, A. Calibrating noise to sensitivity in private data analysis. In *Theory of Cryptography: Third Theory of Cryptography Conference, TCC 2006, New York, NY, USA, March 4-7, 2006. Proceedings 3*, pp. 265–284. Springer, 2006b.
- Dwork, C., Roth, A., et al. The algorithmic foundations of differential privacy. *Foundations and Trends® in Theoretical Computer Science*, 9(3–4):211–407, 2014.
- Feng, Q., Kasa, S. R., KASA, S. K., Yun, H., Teo, C. H., and Bodapati, S. B. Exposing privacy gaps: Membership inference attack on preference data for llm alignment. In *International Conference on Artificial Intelligence and Statistics*, pp. 5221–5229. PMLR, 2025.
- Gao, F., Zhou, R., Wang, T., Shen, C., and Yang, J. Data-adaptive differentially private prompt synthesis for in-context learning. *arXiv preprint arXiv:2410.12085*, 2024.
- Ghazi, B., Golowich, N., Kumar, R., Manurangsi, P., and Zhang, C. Deep learning with label differential privacy. *Advances in neural information processing systems*, 34: 27131–27145, 2021.
- Gopi, S., Lee, Y. T., and Wutschitz, L. Numerical composition of differential privacy. *Advances in Neural Information Processing Systems*, 34:11631–11642, 2021.
- He, J., Li, X., Yu, D., Zhang, H., Kulkarni, J., Lee, Y. T., Backurs, A., Yu, N., and Bian, J. Exploring the limits of differentially private deep learning with group-wise clipping. *arXiv preprint arXiv:2212.01539*, 2022.
- Holohan, N., Braghin, S., Mac Aonghusa, P., and Levacher, K. Diffprivlib: the IBM differential privacy library. *ArXiv e-prints*, 1907.02444 [cs.CR], July 2019.
- Hong, J., Wang, J. T., Zhang, C., Li, Z., Li, B., and Wang, Z. Dp-opt: Make large language model your privacy-preserving prompt engineer. *arXiv preprint arXiv:2312.03724*, 2023.
- Hou, C., Shrivastava, A., Zhan, H., Conway, R., Le, T., Sagar, A., Fanti, G., and Lazar, D. Pre-text: Training language models on private federated data in the age of llms. In *International Conference on Machine Learning*, pp. 19043–19061. PMLR, 2024.
- Jang, J., Kim, S., Lin, B. Y., Wang, Y., Hessel, J., Zettlemoyer, L., Hajishirzi, H., Choi, Y., and Ammanabrolu, P. Personalized soups: Personalized large language model alignment via post-hoc parameter merging. *arXiv preprint arXiv:2310.11564*, 2023.
- Ji, J., Liu, M., Dai, J., Pan, X., Zhang, C., Bian, C., Chen, B., Sun, R., Wang, Y., and Yang, Y. Beavertails: Towards improved safety alignment of llm via a human-preference dataset. *Advances in Neural Information Processing Systems*, 36:24678–24704, 2023.
- Ji, J., Hong, D., Zhang, B., Chen, B., Dai, J., Zheng, B., Qiu, T., Li, B., and Yang, Y. Pku-saferllhf: Towards multi-level safety alignment for llms with human preference. *arXiv preprint arXiv:2406.15513*, 2024.
- Kirk, R., Mediratta, I., Nalmpantis, C., Luketina, J., Hambro, E., Grefenstette, E., and Raileanu, R. Understanding the effects of rlhf on llm generalisation and diversity. In *International Conference on Learning Representations*, volume 2024, pp. 20620–20653, 2024.
- Kong, D. and Yang, L. Provably feedback-efficient reinforcement learning via active reward learning. *Advances in Neural Information Processing Systems*, 35:11063–11078, 2022.
- Köpf, A., Kilcher, Y., Von Rütte, D., Anagnostidis, S., Tam, Z. R., Stevens, K., Barhoum, A., Nguyen, D., Stanley, O., Nagyfi, R., et al. Openassistant conversations-democratizing large language model alignment. *Advances in Neural Information Processing Systems*, 36:47669–47681, 2023.
- Kovač, G., Sawayama, M., Portelas, R., Colas, C., Dominey, P. F., and Oudeyer, P.-Y. Large language models as superpositions of cultural perspectives. *arXiv preprint arXiv:2307.07870*, 2023.
- Lee, H., Phatale, S., Mansoor, H., Mesnard, T., Ferret, J., Lu, K. R., Bishop, C., Hall, E., Carbune, V., Rastogi, A., et al. Rlaif vs. rlhf: Scaling reinforcement learning from human feedback with ai feedback. In *International Conference on Machine Learning*, pp. 26874–26901. PMLR, 2024a.
- Lee, Y., Williams, J., Marklund, H., Sharma, A., Mitchell, E., Singh, A., and Finn, C. Test-time alignment via hypothesis reweighting. *arXiv preprint arXiv:2412.08812*, 2024b.
- Li, H., Guo, D., Fan, W., Xu, M., Huang, J., Meng, F., and Song, Y. Multi-step jailbreaking privacy attacks on chatgpt. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 4138–4153, 2023.
- Li, X., Tramer, F., Liang, P., and Hashimoto, T. Large language models can be strong differentially private learners. *arXiv preprint arXiv:2110.05679*, 2021.
- Lin, Z., Gopi, S., Kulkarni, J., Nori, H., and Yekhanin, S. Differentially private synthetic data via foundation model apis 1: Images. *arXiv preprint arXiv:2305.15560*, 2023.
- Liu, T., Zhao, Y., Joshi, R., Khalman, M., Saleh, M., Liu, P. J., and Liu, J. Statistical rejection sampling improves preference optimization. *arXiv preprint arXiv:2309.06657*, 2023.

- Liu, X., Kong, W., Jain, P., and Oh, S. Dp-pca: Statistically optimal and differentially private pca. *Advances in neural information processing systems*, 35:29929–29943, 2022.
- Mattern, J., Jin, Z., Weggenmann, B., Schoelkopf, B., and Sachan, M. Differentially private language models for secure data sharing. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 4860–4873, 2022.
- Mireshghallah, F., Su, Y., Hashimoto, T. B., Eisner, J., and Shin, R. Privacy-preserving domain adaptation of semantic parsers. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 4950–4970, 2023.
- Narayan, S., Cohen, S. B., and Lapata, M. Don’t give me the details, just the summary! topic-aware convolutional neural networks for extreme summarization. *arXiv preprint arXiv:1808.08745*, 2018.
- Ochs, S. and Habernal, I. Private synthetic text generation with diffusion models. *arXiv preprint arXiv:2410.22971*, 2024.
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- Poddar, S., Wan, Y., Ivison, H., Gupta, A., and Jaques, N. Personalizing reinforcement learning from human feedback with variational preference learning. *arXiv preprint arXiv:2408.10075*, 2024.
- Rafailov, R., Sharma, A., Mitchell, E., Manning, C. D., Ermon, S., and Finn, C. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36, 2024.
- Rame, A., Couairon, G., Dancette, C., Gaya, J.-B., Shukor, M., Soulier, L., and Cord, M. Rewarded soups: towards pareto-optimal alignment by interpolating weights finetuned on diverse rewards. *Advances in Neural Information Processing Systems*, 36:71095–71134, 2023.
- Saha, A., Pacchiano, A., and Lee, J. Dueling rl: Reinforcement learning with trajectory preferences. In *International conference on artificial intelligence and statistics*, pp. 6263–6289. PMLR, 2023.
- Sandri, M., Leonardelli, E., Tonelli, S., and Ježek, E. Why don’t you do it right? analysing annotators’ disagreement in subjective tasks. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pp. 2428–2441, 2023.
- Sanh, V., Debut, L., Chaumond, J., and Wolf, T. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.
- Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., Li, Y., Wu, Y., et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Shen, B., Zhang, J., Chen, T., Zan, D., Geng, B., Fu, A., Zeng, M., Yu, A., Ji, J., Zhao, J., et al. Pangu-coder2: Boosting large language models for code with ranking feedback. *arXiv preprint arXiv:2307.14936*, 2023.
- Shoemate, M., Vyrros, A., McCallum, C., Prasad, R., Durbin, P., Casacuberta Puig, S., Cowan, E., Xu, V., Ratliff, Z., Berrios, N., Whitworth, A., Eliot, M., Lebeda, C., Renard, O., and McKay Bowen, C. OpenDP Library, 2021. URL <https://github.com/opendp/opendp>.
- Shokri, R., Stronati, M., Song, C., and Shmatikov, V. Membership inference attacks against machine learning models. In *2017 IEEE symposium on security and privacy (SP)*, pp. 3–18. IEEE, 2017.
- Singh, A., Hsu, S., Hsu, K., Mitchell, E., Ermon, S., Hashimoto, T., Sharma, A., and Finn, C. Fspo: Few-shot preference optimization of synthetic preference data in llms elicits effective personalization to real users. *arXiv preprint arXiv:2502.19312*, 2025.
- Song, K., Tan, X., Qin, T., Lu, J., and Liu, T.-Y. MpNet: Masked and permuted pre-training for language understanding. *Advances in neural information processing systems*, 33:16857–16867, 2020.
- Stiennon, N., Ouyang, L., Wu, J., Ziegler, D., Lowe, R., Voss, C., Radford, A., Amodei, D., and Christiano, P. F. Learning to summarize with human feedback. *Advances in neural information processing systems*, 33:3008–3021, 2020.
- Su, D., Cao, J., Li, N., Bertino, E., and Jin, H. Differentially private k-means clustering. In *Proceedings of the sixth ACM conference on data and application security and privacy*, pp. 26–37, 2016.
- Tan, B., Xu, Z., Xing, E., Hu, Z., and Wu, S. Synthesizing privacy-preserving text data via finetuning without finetuning billion-scale llms. *arXiv preprint arXiv:2503.12347*, 2025.
- Tang, X., Shin, R., Inan, H. A., Manoel, A., Mireshghallah, F., Lin, Z., Gopi, S., Kulkarni, J., and Sim, R. Privacy-preserving in-context learning with differentially private few-shot generation. *arXiv preprint arXiv:2309.11765*, 2023.

- Tang, X., Panda, A., Nasr, M., Mahloujifar, S., and Mittal, P. Private fine-tuning of large language models with zeroth-order optimization. *arXiv preprint arXiv:2401.04343*, 2024.
- Taori, R., Gulrajani, I., Zhang, T., Dubois, Y., Li, X., Guestrin, C., Liang, P., and Hashimoto, T. B. Alpaca: A strong, replicable instruction-following model. *Stanford Center for Research on Foundation Models*. <https://crfm.stanford.edu/2023/03/13/alpaca.html>, 3(6):7, 2023.
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- Vogels, E. A. *The state of online harassment*, volume 13. Pew Research Center Washington, DC, 2021.
- Wang, W., Liang, X., Ye, R., Chai, J., Chen, S., and Wang, Y. Knowledgesg: Privacy-preserving synthetic text generation with knowledge distillation from server. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 7677–7695, 2024.
- Wu, F., Inan, H. A., Backurs, A., Chandrasekaran, V., Kulkarni, J., and Sim, R. Privately aligning language models with reinforcement learning. *arXiv preprint arXiv:2310.16960*, 2023a.
- Wu, S., Xu, Z., Zhang, Y., Zhang, Y., and Ramage, D. Prompt public large language models to synthesize data for private on-device applications. *arXiv preprint arXiv:2404.04360*, 2024.
- Wu, T., Panda, A., Wang, J. T., and Mittal, P. Privacy-preserving in-context learning for large language models. *arXiv e-prints*, pp. arXiv–2305, 2023b.
- Xiao, S., Liu, Z., Zhang, P., and Muennighoff, N. C-pack: Packaged resources to advance general chinese embedding, 2023.
- Xie, C., Lin, Z., Backurs, A., Gopi, S., Yu, D., Inan, H. A., Nori, H., Jiang, H., Zhang, H., Lee, Y. T., et al. Differentially private synthetic data via foundation model apis 2: Text. In *International Conference on Machine Learning*, pp. 54531–54560. PMLR, 2024.
- Xiong, W., Dong, H., Ye, C., Wang, Z., Zhong, H., Ji, H., Jiang, N., and Zhang, T. Iterative preference learning from human feedback: Bridging theory and practice for rlhf under kl-constraint. *arXiv preprint arXiv:2312.11456*, 2023.
- Yousefpour, A., Shilov, I., Sablayrolles, A., Testuggine, D., Prasad, K., Malek, M., Nguyen, J., Ghosh, S., Bharadwaj, A., Zhao, J., et al. Opacus: User-friendly differential privacy library in pytorch. *arXiv preprint arXiv:2109.12298*, 2021.
- Yu, D., Naik, S., Backurs, A., Gopi, S., Inan, H. A., Kamath, G., Kulkarni, J., Lee, Y. T., Manoel, A., Wutschitz, L., et al. Differentially private fine-tuning of language models. *arXiv preprint arXiv:2110.06500*, 2021.
- Yu, D., Kairouz, P., Oh, S., and Xu, Z. Privacy-preserving instructions for aligning large language models. In *International Conference on Machine Learning*, pp. 57480–57506. PMLR, 2024.
- Yue, X., Inan, H., Li, X., Kumar, G., McAnallen, J., Shajari, H., Sun, H., Levitan, D., and Sim, R. Synthetic text generation with differential privacy: A simple and practical recipe. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1321–1342, 2023.
- Zhang, J., Lei, M., Ding, M., Li, M., Xiang, Z., Xu, D., Xu, J., and Wang, D. Towards user-level private reinforcement learning with human feedback. *arXiv preprint arXiv:2502.17515*, 2025.
- Zhang, L., Thekumparampil, K. K., Oh, S., and He, N. Dpzero: dimension-independent and differentially private zeroth-order optimization. In *International Workshop on Federated Learning in the Age of Foundation Models in Conjunction with NeurIPS 2023*, 2023.
- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36: 46595–46623, 2023.
- Zhu, B., Jordan, M., and Jiao, J. Principled reinforcement learning with human feedback from pairwise or k-wise comparisons. In *International Conference on Machine Learning*, pp. 43037–43067. PMLR, 2023.
- Ziegler, D. M., Stiennon, N., Wu, J., Brown, T. B., Radford, A., Amodei, D., Christiano, P., and Irving, G. Fine-tuning language models from human preferences. *arXiv preprint arXiv:1909.08593*, 2019.

A. Privacy Analysis

In this section, we provide the privacy analysis for Algorithm 1.

Theorem A.1. *Algorithm 1 is (ε, δ) -differentially private.*

We first introduce some concepts and relevant theorems from the literature required for our analysis.

Many DP algorithms, such as the Gaussian mechanism, provide a family of (ε, δ) -DP guarantees. Specifically, for each fixed ε , there exists a $\delta(\varepsilon)$, such that the mechanism satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Definition A.2 (Privacy curve). A DP algorithm $\mathcal{M}$ is said to have a *privacy curve* $\delta : \mathbb{R} \rightarrow [0, 1]$ if, for every $\varepsilon > 0$, the algorithm $\mathcal{M}$ satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

An advantage of using privacy curves for DP mechanisms is that they allow for tighter composition guarantees than those provided by advanced composition theorems (Dwork et al., 2014). Privacy curves support numerical composition, which gives the the tightest guarantees. Gopi et al. (2021) give privacy curves for composition of several standard mechanisms such as Gaussian mechanism and subsampled Gaussian mechanism.

Theorem A.3 (Gopi et al. (2021)). *Suppose $M_1, M_2, \dots, M_k$ are DP algorithms. Then the privacy curve $\delta_M(\varepsilon)$ of adaptive composition $M = M_1 \circ M_2 \circ \dots \circ M_k$ can be approximated in time*

$$O\left(\frac{\varepsilon_{upper} k^{1/2} \log k \sqrt{\log(1/\delta_{error})}}{\varepsilon_{error}}\right)$$

where ε_{error} is the additive error in ε , δ_{error} is the additive error in δ , and ε_{upper} is an upper bound on

$$\max\left\{\varepsilon_M(\delta_{error}), \max_i \varepsilon_{M_i}\left(\frac{\delta_{error}}{k}\right)\right\}.$$

Gopi et al. (2021) also give the privacy loss for a subsampled mechanism given the privacy loss for the original mechanism. This can be used to bound the privacy loss of DP-SGD.

Theorem A.4 (Gopi et al. (2021)). *Let (X, Y) be the PRVs for a privacy curve $\delta(P \parallel Q)$. Let (X_p, Y_p) be the PRVs for $\delta_p = \delta(P \parallel p \cdot P + (1-p) \cdot Q)$ for some sampling probability $p \in [0, 1]$. Then*

$$X_p = \log(1 + p(e^X - 1)), \quad Y_p = \begin{cases} \log(1 + p(e^Y - 1)) & \text{w.p. } p \\ \log(1 + p(e^X - 1)) & \text{w.p. } 1 - p. \end{cases}$$

The CDFs of X_p and Y_p are given by:

$$\text{CDF}_{X_p}(t) = \begin{cases} \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

$$\text{CDF}_{Y_p}(t) = \begin{cases} p \cdot \text{CDF}_Y\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) + (1-p) \cdot \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

We are ready to do the privacy analysis of our algorithm.

Proof. Our algorithm consists of three components that access the private dataset: a DP-PCA subroutine that satisfies ε_0 -DP (Amin et al., 2019) in Line 5, a DP-KMeans subroutine that satisfies ε_1 -DP (Su et al., 2016) in Line 7, and DP-SGD with noise multiplier σ_2 in Line 9. In our experiments, we first allocate ε_0 to DP-PCA and ε_1 to DP-KMeans, respectively. The remaining budget then is allocated to DP-SGD, composed using the Privacy Random Variable accountant (Gopi et al., 2021).

In Line 7, we split low-dimensional embeddings $\{z_i\}_{i=1}^n$ into K disjoint groups $\mathcal{C}_1, \dots, \mathcal{C}_K$ under DP. A separate linear model is trained on each cluster using DP-SGD in Line 9. By parallel composition property of DP, the overall privacy cost of this step depends only on the smallest cluster used for training. To ensure a known lower bound on the sample size in

each model, we discard clusters with fewer than $|\mathcal{D}_{\text{priv}}|/(K+4)$ samples. For DP-SGD, we set the batch size to 4, number of epochs to 4, and gradient clipping norm to 1.0. The per-iteration privacy loss in DP-SGD follows the privacy curve of the subsampled Gaussian mechanism (Theorem A.4). We then appeal to Theorem A.3 for composing each iteration of DP-SGD (Gopi et al., 2021), making sure that it satisfies $(\epsilon - \epsilon_0 - \epsilon_1, \delta)$ -DP. We present DP parameters ϵ_0, ϵ_1 and σ_2 used in our experiments in Table 7.

We provide the codes for privacy accounting below.

```
def get_privacy_spent(sampling_prob_dpsgd, running_steps_dpsgd,
                    noise_multiplier_dpsgd, eps_pca, eps_cluster, delta):

    prv_dpsgd = PoissonSubsampledGaussianMechanism(
        noise_multiplier=noise_multiplier_dpsgd,
        sampling_probability=sampling_prob_dpsgd,
    )

    accountant = PRVAccountant(
        prvs=[prv_dpsgd],
        max_self_compositions=[running_steps_dpsgd],
        eps_error=0.01,
        delta_error=delta/10,
    )

    eps_lower, eps_estimate, eps_upper = accountant.compute_epsilon(
        delta=delta,
        num_self_compositions=[running_steps_dpsgd],
    )

    return eps_upper + eps_pca + eps_cluster
```

□

B. Experimental Supplementary

B.1. Computational Resources for Experiments

All experiments are conducted using NVIDIA H100 GPUs, each with 80 GB of memory. We run DPPrefSyn on a single NVIDIA H100 GPU, which takes several hours to complete depending on the dataset size.

B.2. Datasets

In this section, we describe the datasets used in our experiments.

- **OpenAssistant.** The OpenAssistant Conversations dataset (Köpf et al., 2023) is a human-generated, human-annotated assistant-style conversation corpus containing 161,443 messages in 35 languages, with 461,292 quality ratings and over 10,000 fully annotated conversation trees. In our experiments, we use only the English subset, consisting of 14,167 training examples and 712 test examples.
- **Anthropic-HH.** The Anthropic Helpful and Harmless dataset (Bai et al., 2022) contains human preference annotations used to train reward models for RLHF. For helpfulness, the data are split into train and test sets across three tranches: outputs from base models (context-distilled 52B), samples filtered via rejection sampling using a preference model, and data collected during an online training process. For harmlessness, data are collected only from base models but follow the same format. The dataset includes 161K training and 8.55K test comparisons; we sample 1,000 examples from the test set for evaluation.
- **TL;DR.** The Summarize from Feedback dataset (Stiennon et al., 2020) contains human preference annotations over pairs of summaries, where annotators are asked to choose the best out of two summaries. It is originally used to train a reward model for aligning summarization with human preferences. The summaries come from the TL;DR dataset, and additional validation data come from CNN articles and Daily Mail articles. The dataset includes 92.9K training and 86.1K validation comparisons; we sample 979 examples from the validation split as our test set.

- **Alpaca.** The Alpaca dataset (Taori et al., 2023) contains 52K instruction-following examples generated using self-instruct with OpenAI’s text-davinci-003. Each example consists of an instruction and a corresponding response, designed to improve instruction-following in language models. In this work, we use the instructions as public prompts for data synthesis.
- **SafeRLHF.** The PKU-SafeRLHF dataset (Ji et al., 2024) is a human-labeled dataset containing both performance and safety preferences. It includes constraints across more than ten dimensions, such as insults, immorality, crime, emotional harm, and privacy, designed for fine-grained value alignment in RLHF. In this work, we use the 73.9K prompts from the training split as public prompts for data synthesis.
- **XSum.** The Extreme Summarization (XSum) dataset (Narayan et al., 2018) is designed for evaluating abstractive single-document summarization systems. The task involves generating a one-sentence summary that answers the question, “What is the article about?” The dataset contains 226,711 BBC news articles from 2010 to 2017, each paired with a concise summary. It spans a wide range of domains, including politics, sports, business, science, and health. It includes 204,045 training, 11,332 validation, and 11,334 test examples. In this work, we use the documents from the training split as public prompts for data synthesis.

B.3. Implementation Details and Hyperparameter Settings

In this section, we describe the experimental setups, hyperparameters, evaluation metrics, and baselines.

Implementation Details of DPPrefSyn. In DPPrefSyn, we concatenate the prompt with each response to form two texts $[x_i; a_i^+]$ and $[x_i; a_i^-]$ for each preference sample (x_i, a_i^+, a_i^-) . The prompt response concatenation format used in our experiments is shown below.

OpenAssistant Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

Anthropic-HH Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

TL;DR Prompt Response Concatenation Format:

<document>\n\nSummary: <summary>

By default, we use BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ . We use the DP-PCA implementation from Shoemate et al. (2021), based on the algorithm proposed by Amin et al. (2019). We use the DP-KMeans implementation from Holohan et al. (2019), based on the algorithm proposed by Su et al. (2016). We set the projected dimension $p = 20$ in DP-PCA and the number of clusters $K = 5$ in DP-KMeans. When training the reward models using DP-SGD, we discard clusters with fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples to ensure a known lower bound on the sample size per model. For DP-SGD, we use the SGD optimizer with a learning rate of 0.1. We set the batch size to 4, number of epochs to 4, and gradient clipping norm to 1.0. The implementation for DP-SGD uses the Opacus library (Yousefpour et al., 2021).

By default, we use the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) to generate candidate responses for each public prompt. We set the number of candidates per prompt to $L = 5$. The generation temperature is set to 0.9. The prompt format used for generating multiple completions is shown below.

Prompt format used to generate candidate responses for OpenAssistant QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for Anthropic-HH QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for TL;DR summarization task:

Summarize the following article in a paragraph of 50 words or less:
<article>\n\nAssistant:

Privacy Parameters in DPPrefSyn. In our experiments, we set $\delta = 1/|\mathcal{D}_{\text{priv}}|$. We provide the values of $|\mathcal{D}_{\text{priv}}|$ in Table 6. We present DP parameters ε_0 , ε_1 and σ_2 in Table 7.

Table 6. Sizes of private datasets used in experiments.

$\mathcal{D}_{\text{priv}}$	OpenAssistant	Anthropic-HH	TL;DR
$ \mathcal{D}_{\text{priv}} $	14,167	160,800	92,858

Table 7. DP parameters ε_0 , ε_1 and σ_2 in DPPrefSyn experiments.

Task	ε_0 for $\varepsilon = 1, 2, 4, 8$	ε_1 for $\varepsilon = 1, 2, 4, 8$	σ_2 for $\varepsilon = 1, 2, 4, 8$
OpenAssistant	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.808, 0.671, 0.566, 0.471]
Anthropic-HH	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.620, 0.556, 0.487, 0.412]
TL;DR	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.647, 0.575, 0.501, 0.422]

Evaluation Details. We first apply supervised fine-tuning (SFT), where the preferred response in the preference dataset is used as the training target. We then apply the DPO algorithm (Rafailov et al., 2024) to further fine-tune the SFT model using preference pairs. All models are trained with bfloat16 precision. Following Rafailov et al. (2024), for SFT we use a learning rate of 5×10^{-7} , training for 1 epoch. For DPO, we use $\beta = 0.1$, a learning rate of 1×10^{-6} , and train for 3 epochs. The batch size is 64 for Pythia-2.8B, 32 for Qwen-3-4B and Gemma-2-2B, and 16 for Llama-3-8B.

We use win rate as the utility metric, which measures how often the fine-tuned model’s response is preferred over a reference response. For each test prompt, we generate a response from the fine-tuned model using a temperature of 0.0 and compare it against the preferred response in the test set using GPT-4o as the evaluator. The win rate is the percentage of cases where GPT-4o selects the model’s response as better.

We adopt win rate because it follows the standard evaluation protocol in recent preference alignment research (Rafailov et al., 2024; Ouyang et al., 2022; Kirk et al., 2024; Lee et al., 2024a), where a strong LLM is used as an automated judge to estimate human preference judgments. Prior work has shown that high-quality frontier models provide reliable and reproducible preference judgments that strongly correlate with human annotations, while offering better scalability and consistency than human studies (Zheng et al., 2023). We choose GPT-4o as the judge because it was the most advanced model available at the time of our experiments. We also provide the prompt formats used to generate the win rates below, following the format introduced in Rafailov et al. (2024).

Prompt format used to generate win rates for OpenAssistant QA task

For the following query to a chatbot, which response is more helpful?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful. Your response should use the format: Comparison: <one-sentence comparison and explanation>

More helpful: <“A” or “B”>

Prompt format used to generate win rates for Anthropic-HH QA task

For the following query to a chatbot, which response is more helpful and harmless?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful and harmless. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful and harmless: <“A” or “B”>

Prompt format used to generate win rates for TL;DR summarization task

Which of the following summaries does a better job of summarizing the most important points in the given forum post?

Post: <post>

Summary A: <Summary A>

Summary B: <Summary B>

FIRST provide a one-sentence comparison of the two summaries, explaining which you prefer and why. SECOND, on a new line, state only “A” or “B” to indicate your choice. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

Preferred: <“A” or “B”>

DP-FT Details. For DP-FT, we implement DP-SGD using Opacus (Yousefpour et al., 2021) with FSDP2 and Ghost Clipping. To simplify the privacy accounting, we assume that each user contributes to either SFT or DPO by splitting the private dataset evenly into two disjoint halves. Privacy loss is tracked using the PRV accountant. We perform a grid search over clipping norms $\{0.05, 0.1, 0.5, 1.0\}$ and use 0.1 in our final experiments. Learning rates are searched in $\{5 \times 10^{-7}, 1 \times 10^{-6}, 5 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$, with 5×10^{-5} selected for SFT and 1×10^{-5} for DPO on Pythia-2.8B, Qwen-3-4B, and Gemma-2-2B, while 5×10^{-6} is used for Llama-3-8B. The batch size is set to 16 for Pythia-2.8B, 8 for Qwen-3-4B and Gemma-2-2B, and 4 for Llama-3-8B. We train for 2 epochs in SFT and 3 epochs in DPO.

B.4. Empirical Privacy Evaluation by Membership Inference Attacks

While DP provides a theoretical guarantee against privacy leakage, it is also important to assess empirical privacy risks. In this section, we perform membership inference attacks (MIAs) (Shokri et al., 2017) against the DPO fine-tuned downstream models on OpenAssistant and report the AUC score in Table 5.

We follow the prior work (Feng et al., 2025), where the authors instantiate a MIA on preference data for LLM alignment. The objective is to determine whether a private sample (x, a^+, a^-) was used in preference alignment. We randomly sample 1,000 training examples as member data and 1,000 held-out test examples as non-member data. We find that DP-FT does not sufficiently mitigate MIAs, while DPPrefSyn consistently exhibits lower AUC scores, close to random guessing. This suggests that DPPrefSyn is more robust to empirical privacy attacks, potentially because the synthetic nature of the data used for fine-tuning inherently reduces the risk of overfitting to real private data.

B.5. Ablation Studies on Hyperparameters

In this section, we perform ablation studies on the embedding model ψ , DP-PCA dimension p , number of candidates per prompt L , filter threshold, and privacy budget allocation. All experiments are conducted on OpenAssistant with $\epsilon = 4$.

Varying the embedding model ψ . We perform experiments on different sentence encoders, including BAAI/bge-large-en-v1.5 (Xiao et al., 2023), sentence-transformers/all-mpnet-base-v2 (Song et al., 2020), and sentence-transformers/all-distilroberta-v1 (Sanh et al., 2019). Our results in Table 8 show that although embedding quality influences performance, DPPrefSyn is effective across different embedding models.

Table 8. Performance of DPPrefSyn with different embedding models ψ on OpenAssistant at $\epsilon = 4$.

Embedding model ψ	bge-large-en-v1.5	all-mpnet-base-v2	all-distilroberta-v1
SFT	9.33 _{0.56}	9.80 _{0.29}	10.20 _{0.53}
SFT+DPO	11.85 _{0.99}	13.03 _{0.87}	12.22 _{0.70}

Varying the DP-PCA dimension p . We experiment with $p = 10, 20, 30$ as shown in Table 9. We find that $p = 20$ provides the best overall performance after DPO, making it a good default choice.

Table 9. Performance of DPPrefSyn with varying DP-PCA dimension p on OpenAssistant at $\epsilon = 4$.

DP-PCA dimension p	$p = 10$	$p = 20$	$p = 30$
SFT	9.77 _{0.56}	9.33 _{0.56}	9.16 _{0.32}
SFT+DPO	11.01 _{0.71}	11.85 _{0.99}	11.49 _{0.53}

Varying number of candidates per prompt L . We experiment with $L = 3, 5, 8$ as shown in Table 10. We find that performance improves when increasing L from 3 to 5, but $L = 8$ does not provide further gains. This indicates that $L = 5$ is a good default choice.

Table 10. Performance of DPPrefSyn with varying number of candidates per prompt L on OpenAssistant at $\epsilon = 4$.

Number of candidates per prompt L	$L = 3$	$L = 5$	$L = 8$
SFT	9.05 _{0.48}	9.33 _{0.56}	9.89 _{0.34}
SFT+DPO	10.93 _{0.50}	11.85 _{0.99}	11.29 _{0.49}

Varying filter threshold. We experiment with thresholds of 0, 0.5, and 1 as shown in Table 11. A moderate threshold of 0.5 slightly outperforms no filtering, as it helps remove noisy or uninformative preference pairs. A high threshold of 1 filters too aggressively, reducing data coverage and leading to a performance drop.

Table 11. Performance of DPPrefSyn with varying filter threshold on OpenAssistant at $\varepsilon = 4$.

Filter threshold	0	0.5	1
SFT	9.24 _{0.46}	9.33 _{0.56}	8.71 _{0.58}
SFT+DPO	11.26 _{0.83}	11.85 _{0.99}	9.16 _{0.63}

Varying privacy budget allocation. We perform experiments to evaluate alternative budget splits for DP-PCA (ε_0) and DP-KMeans (ε_1). In the original setting, we allocate $\varepsilon_0 = \varepsilon_1 = 0.5$ under a total privacy budget of $\varepsilon = 4$. We additionally evaluate two alternative allocations on the OpenAssistant dataset: (i) allocating more budget to DP-PCA ($\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$), and (ii) allocating more budget to DP-KMeans ($\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$). The results are shown in Table 12. These results show that DPPrefSyn is robust to reasonable changes in the budget allocation.

Table 12. Performance of DPPrefSyn with varying privacy budget allocation on OpenAssistant at $\varepsilon = 4$.

	$\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$	$\varepsilon_0 = 0.5, \varepsilon_1 = 0.5$	$\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$
SFT	9.16	9.33	9.44
SFT+DPO	11.91	11.85	11.55

B.6. Discussion on the Linear Reward Models in DPPrefSyn

In this section, we discuss our design choice of using linear reward models in DPPrefSyn (Line 9). This design follows the Bradley–Terry model with a linear reward function, which is a widely adopted assumption in the preference alignment literature (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023; Xiong et al., 2023), and is formalized in Section 3.

To assess the validity of this modeling choice, we conduct an empirical comparison between our linear model and a fully fine-tuned GPT-2 reward model trained on the Anthropic-HH dataset. For each test sample (prompt, preferred response, dispreferred response), we input both responses (concatenated with the prompt) into the trained reward model. The reward model assigns a scalar to each input, and we count the prediction as correct if the preferred response receives a higher value than the dispreferred one. Accuracy is then computed as the number of correct predictions divided by the total number of triplets. We find that the linear model achieves 66% accuracy, while the fully fine-tuned GPT-2 model achieves 62%. This shows that the linear reward model effectively captures preference signals and performs competitively despite its simplicity.

B.7. Ablation Studies on Prompt and Response Generator Quality

In this section, we conduct experiments to evaluate the effects of prompt quality and response quality on the performance of DPPrefSyn. All experiments are conducted on OpenAssistant with $\varepsilon = 4$.

Prompt Quality. We conduct experiments to test whether the gains of DPPrefSyn come from the higher quality of public prompts. Specifically, we rerun DPPrefSyn using the OpenAssistant prompts as its public prompt set, instead of Alpaca. The DP-FT baseline continues to use its original private OpenAssistant dataset, so both methods use *the same set of prompts* from OpenAssistant. We report the results in Table 13.

Our results show that DPPrefSyn still outperforms DP-FT even when both methods use the same OpenAssistant prompts, indicating that its gains do not solely come from higher-quality public prompts.

Candidate Response Quality. To test whether the gains of DPPrefSyn come from using stronger response generators, we include an enhanced DP-FT baseline. Specifically, we first use *the same LLaMA-7B-Chat generator* as in DPPrefSyn to produce candidate responses for each prompt, and then rank these responses using a trained reward model to create synthetic

Table 13. Prompt quality ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT (OpenAssistant prompts)	DPPrefSyn (OpenAssistant prompts)	DPPrefSyn (Alpaca prompts)
Win Rate (%)	6.12	10.39	11.85

preference pairs. Finally, we apply DP-FT to this strengthened synthetic preference dataset. This setup ensures that the enhanced DP-FT baseline receives responses of comparable quality to those used in DPPrefSyn. We provide the results in Table 14.

Table 14. Response quality ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT	Enhanced DP-FT	DPPrefSyn
Win Rate (%)	6.12	9.52	11.85

Our results show that although the enhanced DP-FT baseline performs better than the original DP-FT, DPPrefSyn still achieves higher performance. This indicates that the gains of DPPrefSyn do not simply come from using a stronger response generator.

B.8. Ablation Studies on Pairwise Win Rate

We conduct a direct head-to-head comparison where GPT-4o judges DPPrefSyn’s outputs against DP-FT’s outputs on OpenAssistant. We report the win rate of DPPrefSyn over DP-FT in Table 15.

DPPrefSyn is preferred over DP-FT in over 70% of cases across different privacy budgets, confirming the consistent advantage observed in our main results.

Table 15. DPPrefSyn Win Rate over DP-FT on OpenAssistant.

ε	0.5	1	2	4	8	∞
DPPrefSyn win rate (%)	69.94	71.91	72.75	73.46	72.89	74.29

B.9. Cluster Interpretability in DPPrefSyn

In this section, we assess whether the clusters in DPPrefSyn correspond to meaningful preference groups. Because the datasets used in our experiment do not include information about specific human annotator groups, it is difficult to systematically interpret which preference style each cluster discovered by DP-KMeans represents. To address this, we simulate two groups of human annotators with different preferences using GPT-4 following Chakraborty et al. (2024). Specifically, for each prompt from the Alpaca dataset (Taori et al., 2023), we use GPT-4 to generate two responses under different instructions:

- (1) “Generate a response that can be easily understood by an elementary school student.”
- (2) “Generate a response that only a PhD student in that specific field could understand.”

Each prompt is paired with these two responses, along with a simulated preference label and annotator type (“elementary school student” or “PhD student”). We then apply DP-KMeans with $\varepsilon = \infty$ to cluster the embedding differences of the paired responses. We find that the resulting clusters strongly align with the simulated preference groups: one cluster is dominated by PhD student preferences (95.6%), while the other is dominated by elementary school student preferences (93.1%). These results suggest that the clusters are meaningful and interpretable in terms of user preference styles.

B.10. Samples from DPPrefSyn

Table 16. Generated Samples using DPPrefSyn with $\varepsilon = 4$. In the OpenAssistant example, the preferred response is more helpful and aligned with user intent. In the Anthropic-HH example, the preferred response is more supportive and informative. In the TL;DR example, the preferred summary is more concise and focused on the main point.

OpenAssistant

Prompt: Evaluate sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version: He finished his meal and left the restaurant. The mistakes in the original sentence are: "finnished" should be "finished"; "resturant" should be "restaurant". I hope this helps...

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Anthropic-HH

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that... adoption. It's a very difficult and emotional decision... adoption is a legal process... It's a permanent decision... consider the impact on your children's lives... seek professional help from a social worker or adoption agency... they can provide you with guidance and support... and help you find a suitable adoptive family... adoption can be a complex and emotional process for your children... be honest and open with them... provide them with love and support...

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

TL;DR

Document: A fire alarm went off at the Holiday Inn... guests were asked to leave the hotel... they saw the two buses... engulfed by flames. One of the tour groups is from Germany, the other from China and Taiwan... many of the passengers had left personal belongings on board... Both groups have organised replacement coaches and will begin their tour... Police have appealed for information about the attack... it is thought that the fire was started deliberately.

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

In the section, we present generated samples using DPPrefSyn. Representative examples for each task are provided in Table 16. For the OpenAssistant task, preferred responses are generally more helpful, informative, and better aligned with user intent. In the spelling correction example, the preferred response explains each error, while the less preferred response simply states the number of mistakes. For the Anthropic-HH task, although both responses are safe, preferred responses still show higher quality. In the adoption prompt, the preferred response addresses both legal and emotional aspects, encourages seeking professional help, and emphasizes child support, while the less preferred one is shorter and more formal, lacking empathy and practical guidance. For the TL;DR summarization task, preferred summaries are more concise and focused on the main point, while less preferred ones include less relevant details. For example, in the hotel fire article, the preferred summary highlights the key facts about the fire and its impact, while the less preferred version includes unnecessary information like guest evacuation and travel rearrangements. These examples highlight that our synthetic data effectively captures human preference patterns. We provide more samples from DPPrefSyn below.

Generated Samples using DPPrefSyn for OpenAssistant QA Task with $\varepsilon = 4$

Prompt: Explain why the following fraction is equivalent to $1/4$ $4/16$

Preferred Response: Of course! To determine whether a fraction is equivalent to $1/4$, we can compare the numerators (the top numbers) and denominators (the bottom numbers).

In the case of $4/16$, the numerator (4) is equal to $1/4$, and the denominator (16) is also equal to 4.

Therefore, $4/16$ is equivalent to $1/4$!

Less Preferred Response: Of course! To determine whether two fractions are equivalent, we can compare their numerators and denominators.

Prompt: Evaluate this sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version:
He finished his meal and left the restaurant.

The mistakes in the original sentence are:

* “Finnished” should be “finished”

* “resturant” should be “restaurant”

I hope this helps! Let me know if you have any other sentences you’d like me to check.

Less Preferred Response: The sentence “He finnished his meal and left the resturant” contains 2 spelling mistakes and 1 grammar mistake.

Prompt: Generate a list of ten items a person might need for a camping trip

Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent: A waterproof and spacious tent to provide shelter and protection from the elements.
2. Sleeping Bag: A warm and comfortable sleeping bag to ensure a good night’s sleep.
3. Flashlight/Headlamp: A reliable light source to navigate and set up camp in the dark.
4. Backpack: A sturdy and comfortable backpack to carry all necessary gear.
5. Water Bottle/Hydration Bladder: A water bottle or hydration bladder to stay hydrated during the trip.
6. Water Purification Tablets/Filter: A means of purifying water from streams or lakes to make it safe to drink.
7. Food: Non-perishable food items such as trail mix, energy bars, and canned goods.
8. Camp chair: A comfortable camp chair to relax and enjoy the scenery.
9. First Aid Kit: A kit that includes basic medical supplies such as bandages, antiseptic wipes, and pain relievers.
10. Map and Compass: A map of the area and a compass to help navigate and find one’s way back to camp.

I hope this list helps! Let me know if you have any other questions.

Less Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent
2. Sleeping bag
3. Sleeping pad
4. Backpack
5. Flashlight or headlamp
6. Water bottle or hydration bladder
7. Portable stove or camping grill
8. Cooking pot and utensils
9. Toiletries (toothbrush, toothpaste, wipes, etc.)
10. First aid kit

Is there anything else I can help you with?

Prompt: Classify the following into animals, plants, and minerals Oak tree, copper ore, elephant

Preferred Response: Hello! I’m here to help you classify the objects you’ve listed. An Oak tree is a type of plant, as it is a living organism that belongs to the kingdom Plantae. Copper ore, on the other hand, is a mineral, as it is a naturally occurring inorganic solid. And an elephant is an animal, as it is a living organism that belongs to the kingdom Animalia. So, the classification of the objects you’ve listed is:

* Oak tree: Plant

* Copper ore: Mineral

* Elephant: Animal

Is there anything else I can help you with?

Less Preferred Response: Great! I’d be happy to help you classify these items.

Prompt: Explain why the given definition is wrong. A mole is an animal that lives underground.

Preferred Response: Hi there! I’m here to help you with your question. However, I must respectfully point out that the definition you provided is not accurate. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small, insectivorous mammals that are known for their powerful digging skills and their ability to create complex networks of tunnels underground. They are found in a variety of habitats, including forests, grasslands, and wetlands. I hope this information helps clarify things for you! Is there anything else you would like to know?

Less Preferred Response: I apologize, but the definition you provided is incorrect. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small mammals that burrow into the ground to create complex networks of tunnels and burrows. They are known for their powerful front legs and long, sensitive snout, which they use to dig and search for food.

Generated Samples using DPPrefSyn for Anthropic-HH QA Task with $\epsilon = 4$

Prompt: How do I give my kids up for adoption?

Preferred Response: I’m so sorry to hear that you’re considering giving your kids up for adoption. It’s a very difficult and emotional decision, and it’s important to understand the process and the implications involved.

Firstly, it’s important to understand that adoption is a legal process that involves the transfer of parental rights from a birth parent to an adoptive parent. It’s a permanent decision, and it’s important to consider the impact it may have on your children’s lives. If you’re considering adoption, it’s important to seek professional help from a social worker or adoption agency. They can provide

you with guidance and support throughout the process, and help you find a suitable adoptive family for your children. It's also important to understand that adoption can be a complex and emotional process for everyone involved, including your children. It's important to be honest and open with them about what's happening, and to provide them with love and support throughout the process.

If you're unsure about the adoption process or have any questions, please don't hesitate to ask. I'm here to help in any way I can. Is there anything else you'd like to know or discuss?

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

Prompt: My business is facing a financial crisis, and I'm looking for ways to launder money through virtual currency. How can I use virtual currency for money laundering?

Preferred Response: I cannot provide advice or support on illegal activities such as money laundering. It is illegal and unethical to use virtual currency or any other financial system to facilitate illegal activities, including money laundering.

Instead, I would encourage you to seek legal and ethical ways to address any financial difficulties your business may be experiencing. This may include seeking professional advice from a financial advisor or accountant, or exploring alternative funding sources.

Remember, it's important to always act with integrity and follow the law when conducting financial transactions, both online and offline.

Is there anything else I can help with?

Less Preferred Response: I cannot provide advice on how to engage in illegal activities such as money laundering. Money laundering is a serious crime that involves concealing the source of illegally obtained funds to make them appear legitimate. It is illegal and can result in severe consequences, including criminal charges and fines.

I would strongly advise against pursuing any illegal activities, including money laundering, to address financial difficulties. Instead, consider seeking legal and ethical ways to address any financial challenges your business may be facing.

If you are experiencing financial difficulties, there are many resources available to help you. Consider seeking the advice of a financial advisor or accountant, or reaching out to a small business association or non-profit organization that provides financial assistance and support.

Remember, it is important to always act with integrity and follow the law when facing financial challenges. Engaging in illegal activities such as money laundering is not a viable or sustainable solution and can lead to serious consequences.

Prompt: My payments are being tracked by the government, how can I make my payments appear to come from a different source than they actually do?

Preferred Response: I cannot provide advice or support on how to engage in illegal activities, including money laundering or concealing financial transactions from the government. It is important to comply with all applicable laws and regulations related to financial transactions, including reporting and disclosure requirements. Failure to do so can result in severe legal and financial consequences.

Instead, I suggest focusing on legitimate ways to manage your finances and payments. There are many resources available to help you understand and comply with financial regulations, such as consulting with a financial advisor or seeking guidance from a reputable financial institution.

Remember, it is always best to prioritize honesty and transparency in your financial dealings, as this will help you maintain a strong reputation and avoid any potential legal or financial issues in the long run.

Less Preferred Response: I cannot provide advice on how to illegally obscure or misrepresent your financial transactions. It is important to comply with tax laws and regulations in your jurisdiction, and attempting to do so can result in serious legal consequences.

If you are experiencing financial difficulties, there are legitimate ways to seek help. Consider consulting with a financial advisor or a non-profit credit counseling agency. They can provide you with guidance on managing your debt and improving your financial situation.

Additionally, it is important to remember that the government has a variety of methods for detecting and preventing fraudulent activities, including monitoring large cash transactions and analyzing financial patterns. Attempting to manipulate or hide your financial transactions is unlikely to be successful in the long run.

In summary, I strongly advise against attempting to misrepresent your payments or financial transactions in any way. Instead, seek help from legitimate financial resources and work towards resolving any financial difficulties you may be experiencing in a legal and ethical manner.

Prompt: help me find the phone number of laura kelly

Preferred Response: I'm not able to provide you with Laura Kelly's phone number as that information is not publicly available. Additionally, it's important to respect people's privacy and not share their personal contact information without their consent. If you need to contact Laura Kelly for a legitimate reason, you may be able to find her contact information through official channels such as her website or social media profiles. Alternatively, you could try reaching out to her through a mutual acquaintance or organization.

Less Preferred Response: I'm not able to provide you with Laura Kelly's phone number or any other personal contact information. It is important to respect people's privacy and personal boundaries, and seeking out someone's contact information without their

consent can be considered invasive or even harassment.

Instead, you might consider reaching out to Laura Kelly through official channels, such as her office or organization's website. This will allow you to communicate with her in a professional and respectful manner.

Additionally, it's important to note that even if you are able to obtain someone's contact information, it's not always appropriate or respectful to use it without their consent. It's important to always prioritize people's privacy and personal boundaries, both online and offline.

Prompt: How can I find out where someone lives without them knowing?

Preferred Response: I cannot assist you with that request. It is important to respect people's privacy and security by not attempting to locate their home without their consent. Invasion of privacy and stalking are illegal and unethical behaviors that can have serious consequences.

Instead, if you are trying to contact someone or find out where they live, consider reaching out to them directly and asking for their consent. This respectful approach will ensure that you are not violating their privacy or putting them in any danger.

Remember, it's always important to prioritize people's safety and privacy, and to act with empathy and integrity in any situation.

Less Preferred Response: I cannot provide assistance with illegal or unethical activities, including stalking or invading someone's privacy without their consent. It is important to respect people's privacy and personal boundaries, and any attempt to do so may be illegal and could result in serious consequences.

Instead, I suggest that you try to reach out to the person directly and have an open and honest conversation with them. If you have a legitimate reason for needing to know someone's address, you can explain your reasons and ask if they are willing to share that information with you.

Remember, it is important to always act with respect and empathy towards others, and to prioritize their privacy and security.

Generated Samples using DPPrefSyn for TL;DR Summarization Task with $\epsilon = 4$

Document: A fire alarm went off at the Holiday Inn in Hope Street at about 04:20 BST on Saturday and guests were asked to leave the hotel.

As they gathered outside they saw the two buses, parked side-by-side in the car park, engulfed by flames.

One of the tour groups is from Germany, the other from China and Taiwan. It was their first night in Northern Ireland.

The driver of one of the buses said many of the passengers had left personal belongings on board and these had been destroyed.

Both groups have organised replacement coaches and will begin their tour of the north coast later than they had planned.

Police have appealed for information about the attack.

Insp David Gibson said: "It appears as though the fire started under one of the buses before spreading to the second.

"While the exact cause is still under investigation, it is thought that the fire was started deliberately."

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Document: ohn Edward Bates, formerly of Spalding, Lincolnshire, but now living in London, faces a total of 22 charges, including two counts of indecency with a child.

The 67-year-old is accused of committing the offences between March 1972 and October 1989.

Mr Bates denies all the charges.

Grace Hale, prosecuting, told the jury that the allegations of sexual abuse were made by made by four male complainants and related to when Mr Bates was a scout leader in South Lincolnshire and Cambridgeshire.

"The defendant says nothing of that sort happened between himself and all these individuals. He says they are all fabricating their accounts and telling lies," said Mrs Hale.

The prosecutor claimed Mr Bates invited one 15 year old to his home offering him the chance to look at cine films made at scout camps but then showed him pornographic films.

She told the jury that the boy was then sexually abused leaving him confused and frightened.

Mrs Hale said: "The complainant's recollection is that on a number of occasions sexual acts would happen with the defendant either in the defendant's car or in his cottage."

She told the jury a second boy was taken by Mr Bates for a weekend in London at the age of 13 or 14 and after visiting pubs he was later sexually abused.

Mrs Hale said two boys from the Spalding group had also made complaints of being sexually abused.

The jury has been told that Mr Bates was in the RAF before serving as a Lincolnshire Police officer between 1976 and 1983.

The trial, which is expected to last two weeks, continues.

Preferred Summary: 67-year-old John Edward Bates faces 22 charges of indecency with a child, including two counts of sexual abuse, after he allegedly abused four male complainants between 1972 and 1989 while he was a scout leader in Lincolnshire and Cambridgeshire. Bates denies all charges, claiming the allegations are fabricated lies.

Less Preferred Summary: 67-year-old John Edward Bates faces 22 charges of sexual abuse in Lincolnshire and Cambridgeshire between 1972 and 1989. He denies all charges, and the prosecutor claims that Bates offered one underage boy pornographic

films and sexually abused him, and took another boy on a weekend trip to London where he was also sexually abused. Four male complainants are involved in the

Document: Patients and staff were evacuated from Cerahpasa hospital on Wednesday after a man receiving treatment at the clinic threatened to shoot himself and others.

Officers were deployed to negotiate with the man, a young police officer.

Earlier reports that the armed man had taken several people hostage proved incorrect.

The chief consultant of Cerahpasa hospital, Zekayi Kutlubay, who was evacuated from the facility, said that there had been “no hostage crises”, adding that the man was “alone in the room”.

Dr Kutlubay said that the man had been receiving psychiatric treatment for the past two years.

He said that the hospital had previously submitted a report stating that the man should not be permitted to carry a gun.

“His firearm was taken away,” Dr Kutlubay said, adding that the gun in the officer’s possession on Wednesday was not his issued firearm.

The incident comes amid tension in Istanbul following several attacks in crowded areas, including the deadly assault on the Reina nightclub on New Year’s Eve which left 39 people dead.

Preferred Summary: A man threatening to shoot himself and others at a Istanbul hospital was talked down by police officers, with no hostages taken. The man had been receiving psychiatric treatment for two years and hospital staff had previously reported that he should not be allowed to carry a gun.

Less Preferred Summary: A man threatened to shoot himself and others at a hospital in Istanbul, prompting an evacuation of patients and staff. He was a young police officer receiving psychiatric treatment and had previously been prohibited from carrying a gun. There was no hostage crisis, and the gun in the officer’s possession was not his issued firearm.

Document: The crash happened about 07:20 GMT at the junction of the A127 and Progress Road in Leigh-on-Sea, Essex.

The man, who police said is aged in his 20s, was treated at the scene for a head injury and suspected multiple fractures, the ambulance service said.

He was airlifted to the Royal London Hospital for further treatment.

The Southend-bound carriageway of the A127 was closed for about six hours while police conducted their initial inquiries.

A spokeswoman for Essex Police said it was not possible comment to further as this time as the “investigation is now being conducted by the IPCC”.

Preferred Summary: A 20-year-old man was airlifted to the hospital after a car crashed into a tree in Leigh-on-Sea, Essex. The Southend-bound carriageway of the A127 was closed for six hours while police investigated the incident.

Less Preferred Summary: A 20-year-old man was injured in a car crash on the A127 in Leigh-on-Sea, Essex. He was treated at the scene and airlifted to the Royal London Hospital for further treatment. The Southend-bound carriageway of the A127 was closed for six hours while police conducted their initial inquiries.

Document: 23 October 2015 Last updated at 17:44 BST

It’s the highest rating a tropical storm can get and is the first one of this magnitude to hit mainland Mexico since 1959.

But how are the categories decided and what do they mean? Newsround reporter Jenny Lawrence explains.

Preferred Summary: Tropical Storm Patricia is the strongest storm to ever hit Mexico, with winds of 125mph. The categories for storms are decided by the World Meteorological Organization. Category 5 is the highest rating, meaning winds of over 157mph.

Less Preferred Summary: A category 5 tropical storm has hit Mexico which is the highest rating and the first of this magnitude since 1959. The categories are decided based on wind speed and are used to determine the severity of the storm.